

Holonomy obstructions for composed local operations, and why they succeed where marginal obstructions fail

Abstract

We record a cohomological obstruction that is non-trivial, computable, robust to noise, and not killed by any of the three theorems that closed the marginal-obstruction programme. The setting is a family of *known* local operations composed along paths—light crossing surfaces in a room, or any system in which each step applies a transformation fixed in advance. A global assignment consistent with every step exists if and only if the holonomy around every loop vanishes, so the obstruction is a class in $H^1(\Gamma, A) \cong A^{\beta_1}$ for a graph Γ and abelian coefficient group A . We verify this computationally against brute force. We then situate it against the two cohomology theories it might be expected to bridge, and state precisely what it does and does not do for that bridge: it shares the coefficient type of Abramsky–Mansfield–Barbosa (a fixed abelian group, not probabilistic functionals) and the locality of Vigneaux (operations attached to a site), but it is a cocycle of *transformations* rather than of sections or of information functions, and therefore is not a bridge but a third object. We list what is proved, what is measured, and what remains open.

1 The construction

Definition 1 (Operation system). Let $\Gamma = (V, E)$ be a finite connected graph, A an abelian group, and $g : E \rightarrow A$ an assignment of a *known* operation to each edge, with $g_{ba} = -g_{ab}$. Call (Γ, A, g) an operation system.

Definition 2 (Global field). A global field is $c : V \rightarrow A$ with $c(b) - c(a) = g_{ab}$ for every edge ab .

Theorem 3 (Holonomy obstruction). *A global field exists if and only if the holonomy $h(L) = \sum_{e \in L} \pm g_e$ vanishes for every cycle L . Equivalently, g is a 1-cocycle on Γ and a global field is a 0-cochain whose coboundary is g ; the obstruction is the class*

$$[g] \in H^1(\Gamma, A) \cong A^{\beta_1}, \quad \beta_1 = |E| - |V| + 1.$$

Proof. Fix a spanning tree T and a root v_0 . Setting $c(v_0) = 0$ and integrating g along tree paths defines c uniquely on V and satisfies the edge condition on T . For a non-tree edge ab , the condition $c(b) - c(a) = g_{ab}$ holds exactly when the holonomy of the fundamental cycle through ab vanishes. The fundamental cycles of T form a basis of the cycle space, of rank β_1 , and the holonomy is linear in g , so vanishing on a basis is vanishing on every cycle. The induced map $g \mapsto (h(L_1), \dots, h(L_{\beta_1}))$ identifies $H^1(\Gamma, A)$ with A^{β_1} . $\square$

This is classical—it is the obstruction to a flat section of a principal bundle over a graph, and for A non-abelian it is the usual non-abelian H^1 of conjugacy classes of holonomy representations. We state it because the point of this note is *which* obstruction survives, not its novelty.

1.1 The smooth form

The same statement in sheaf language, which is where the coefficient group appears most plainly. Let M be a smooth connected manifold with boundary and $r : \partial M \hookrightarrow M$ the inclusion. The exponential sequence of sheaves of abelian groups

$$0 \longrightarrow \mathbb{Z}_M \xrightarrow{i} \mathbb{R}_M \xrightarrow{e} \mathcal{S}_M^1 \longrightarrow 0, \quad e(f) = \exp(2\pi i f), \quad (1)$$

is exact: $\ker e = \mathbb{Z}_M$, and e is surjective on stalks because a local section of $\mathcal{S}_M^1$ admits a smooth logarithm on a contractible neighbourhood.

Theorem 4 (Holonomy as a connecting class). *$\mathbb{R}_M$ is soft, so $H^1(M, \mathbb{R}_M) = 0$ and the connecting map $\delta_M : H^0(M, \mathcal{S}_M^1) \rightarrow H^1(M, \mathbb{Z}_M)$ of (1) is an isomorphism modulo global real lifts. For $[g] \in H^0(M, \mathcal{S}_M^1)$ representing a flat class $\alpha \in H^1(M; \mathbb{R}/\mathbb{Z})$, naturality of the connecting homomorphism under r gives*

$$r^*(\delta_M([g])) = \delta_{\partial M}(r^*[g]) = \text{Hol}(\alpha, \partial M) \in H^1(\partial M, \mathbb{Z}).$$

Proof. Choose a good cover $\{U_i\}$ of M and smooth lifts f_i with $e(f_i) = g|_{U_i}$. On overlaps $c_{ij} = f_i - f_j$ satisfies $e(c_{ij}) = 1$, so $c_{ij} \in \mathbb{Z}_M(U_i \cap U_j)$ and $\{c_{ij}\}$ represents $\delta_M([g])$. Restricting and pairing with a boundary 1-cycle γ gives $\sum_{(i,j) \in \gamma} c_{ij} = \oint_\gamma d\alpha$, the holonomy mod $\mathbb{Z}$. The square commutes by naturality. $\square$

Remark 5 (Two things this theorem does not say). First, $\delta_M([g])$ lies in $H^1(M, \mathbb{Z}_M)$, not in $\text{Ext}^1(\mathcal{S}_M^1, \mathbb{Z}_M)$: the extension class of (1) is a single class independent of g , while the connecting map produces one cohomology class per section, and the two must not be conflated. Second, this class is *not* the AMB obstruction. The tempting identification passes through a “support reduction” $\chi : \mathcal{F}_{\text{prob}} \rightarrow \mathbb{Z}$ sending positive support to 1 and empty support to 0, which is not additive—two positive sections sum to a positive section—hence not a morphism of sheaves of abelian groups, hence not defined on cohomology. This is the same failure as the rounding map and the support indicator of Section 3, in a third costume.

Remark 6 (Verification). `holonomy.py` computes a cycle basis, evaluates the holonomy of each basis loop, and independently constructs a global field by propagation along edges, reporting a contradiction if one appears. On a room with $|V| = 4$, $|E| = 5$, $\beta_1 = 2$ and $A = \mathbb{Z}/4096$, the two agree: holonomy $(0,0)$ with a field existing; $(0,1)$ and $(2,1)$ with no field. The two components vary independently, as A^{β_1} requires.

2 Why this survives where the marginal obstructions did not

Three results closed the earlier programme. None applies here, and the reasons are worth separating because they are different.

Theorem 7 (Single-source global section). *Let $\{P_C\}_{C \in \mathcal{M}}$ be context marginals estimated from one sample $\mathcal{D}$. Then the empirical joint $P_{\mathcal{D}}$ restricts to each P_C , so the family extends; by [1, Prop. 4.3] the obstruction $\gamma(s)$ vanishes for every section and the contextual fraction is 0, on any cover, acyclic or not.*

Proof. Immediate: $P_{\mathcal{D}}$ is a global section by construction. $\square$

Theorem 7 is why no obstruction can be found inside a single corpus at any level or under any cover, and it was confirmed experimentally on cyclic sliding-window covers, where $\gamma = 0$ and $\text{CF} = 0$ for every patch of real and of shuffled data.

Proposition 8 (Theorem 7 does not apply to operation systems). *An operation system has no sample and no marginals. A witness would have to be a global field, which is the very thing whose existence is in question.*

Proposition 9 (Vorob'ev does not apply). *Acyclicity of a cover forces marginal families to extend. Here the relevant structure is the cycle space of Γ : $\beta_1 = 0$ makes H^1 trivial, and $\beta_1 > 0$ leaves it free to be non-zero. The roles of acyclicity coincide, but the objects do not: one is a hypergraph of contexts, the other a graph of paths.*

Proposition 10 (Possibilistic fragility does not apply). *The AMB obstruction is a functional of the support. For an empirical model smoothed at any $\varepsilon > 0$ the support is full, hence possibilistically extendable, hence $\gamma \equiv 0$ while the contextual fraction remains $1 - O(\varepsilon)$; measured at $\varepsilon = 10^{-9}$, γ falls from 9/9 to 0/27. The holonomy class is instead a sum of the operations themselves: it is continuous in g and has no support to blur.*

3 The coefficient group is what mattered

Four constructions were attempted before this one, and all four failed at the same place. The record is worth keeping because the failures were not independent.

attempted coefficients	why it failed	verified
p -adic lattice $\rho : \mathbb{Q} \rightarrow L$	order-preserving + constant on p -adic balls $\Rightarrow$ constant	proof
probability simplex Δ^{V-1}	convex set, no inverses, so no δ	—
support indicator $\omega = \int_{\text{supp}} \mathbf{1} dP$	$\equiv 1$ on every face of every family	computed
rounding $\mathbb{R} \rightarrow \mathbb{Z}$	not additive, so not a module map	—
$\mathbb{Z}/n$ on edges	abelian, δ exists, class varies	computed

Remark 11 (β_1 changes role). In the marginal setting β_1 of the incidence complex is a function of the cover alone: for the 3×3 parity scenario it equals 4 for the obstructed family and for both controls, so it carries no information about the obstruction and cannot be mapped to γ . In an operation system β_1 is the rank of the obstruction group, and the components vary independently with g . The same integer, a different role, because here the edges carry data.

4 The two coboundary operators

The following is an exposition of Baudot–Bennequin [2], included because every statement in Section 5 depends on getting it right, and because four earlier drafts of this note got it wrong in four different ways.

Definition 12 (Conditioning action). For a measurable functional F of a joint distribution P and a discrete variable S_0 ,

$$(S_0 \cdot F)(P) = \sum_j P(S_0 = v_j) F(P \mid S_0 = v_j).$$

For a constant c this gives $S_0 \cdot c = c$, since the weights sum to one.

Definition 13 (The two coboundaries).

$$(\delta F)(S_0; \dots; S_k) = S_0 \cdot F(S_1; \dots; S_k) + \sum_{i=1}^k (-1)^i F(S_0; \dots; S_{i-1} S_i; \dots; S_k) + (-1)^{k+1} F(S_0; \dots; S_{k-1}),$$

and δ_t is the same expression with the first term replaced by $F(S_1; \dots; S_k)$. Both square to zero. The middle terms *merge* adjacent arguments; they do not condition.

Theorem 14 (Baudot–Bennequin).

1. $(\delta c)(S_0) = S_0 \cdot c - c = 0$, so $H^0(\mathcal{F}, \delta) \cong \mathbb{R}$.
2. $(\delta H)(S_0; S_1) = H(S_0) + H(S_1 | S_0) - H(S_0, S_1) = 0$: entropy is a δ -cocycle, by the chain rule. Under locality and measurability hypotheses $H^1(\mathcal{F}, \delta)$ is one-dimensional, generated by H .
3. $(\delta_t H)(S_0; S_1) = H(S_0) + H(S_1) - H(S_0, S_1) = I(S_0; S_1)$, and $(\delta - \delta_t)H = (S_0 - 1) \cdot H(S_1) = -I(S_0; S_1)$. Mutual information is exactly the failure of the module action to be trivial.

Remark 15 (Consequences that are easy to get backwards). $I_2 = \delta_t H$, not δH ; writing the latter is an error, because $\delta H = 0$. More generally the higher mutual informations are coboundaries for δ or for δ_t according as their order is odd or even, so “ I_k is a coboundary” is a statement in the complex selected by parity, not in every complex. And no regularity or symmetry condition separates H from I in a single complex: they are built from the same $p \log p$ terms and have identical boundary behaviour, so any cochain space containing I contains H , whence $I = \delta_t H$ is a coboundary there; while at degree one there is a single argument and no permutation action, so S_k -invariance is vacuous where H lives. The separation is achieved by the module action and by nothing else.

5 Relation to AMB and to Vigneaux

Theorem 16 (Non-unification, [1, 3]). *No morphism of coefficient systems carries a non-zero AMB class in $\mathcal{F}_{\mathbb{Z}}$ to a non-zero class in Vigneaux’s $\mathcal{F}_{\alpha}$.*

Two independent arguments support this; the second is the robust one.

Degree argument, corrected. Baudot and Bennequin show that the higher mutual informations are coboundaries—for δ or for δ_t according as their order is odd or even—and that I_N is an N -cocycle for whichever operator applies. In particular $I_2 = \delta_t H$, not δH : entropy is a δ -cocycle, so $\delta H = 0$. The two operators do not commute, $(\delta - \delta_t)F = ((S_0 - 1) \cdot F)$, so the two complexes have different cohomology. The vanishing of I_k is therefore a statement in the complex selected by its parity, not in every complex, and H^2 of the δ -complex is not computed in [2]—it is their own Problem 1. Any argument resting on universal vanishing is unsound.

Module argument. $\mathcal{F}_{\mathbb{Z}}(C)$ is the free abelian group on support sections; its operation is formal $\mathbb{Z}$ -linear combination and it has no probabilities. $\mathcal{F}_{\alpha}$ assigns real-valued functionals of a probability and its module structure is the conditioning action, which reweights by $P(X=i)$. A morphism would have to intertwine these, and the source has nothing for the action to act on. Proposition 10 is the empirical shadow of this: the two invariants are functions of arguments that vary independently.

Remark 17 (What the holonomy class does for the bridge). It is not a bridge. It shares with AMB the *type* of coefficient—a fixed abelian group, independent of the probabilities—and with Vigneaux the attachment of data to a site. But it is a cocycle of *transformations*, whereas AMB’s is a cocycle of sections and Vigneaux’s of information functions. Three cocycles, three coefficient systems, and the barrier of Theorem 16 is unaffected by the third. What the holonomy construction shows is narrower and more useful: the reason the earlier bridges failed was the coefficient system, not the site, and choosing an honest abelian group makes a non-trivial class appear immediately. The same diagnosis points to Problem 24: if the obstacle is that one coefficient system carries an action the other cannot receive, the repair is not to map between them but to extend one by the other.

6 The three layers, and what passes between them

The results above fall naturally into three levels of realisation. Read forward, they describe a pipeline: an information structure fixes what can be jointly measured, probabilities are then determined on that structure, and outcomes are what those probabilities realise. Read backward, they do not compose: the invariant of each layer is annihilated by the hypotheses of the others, which is the content of Theorem 27 and of Section 5.

layer	invariant	condition for non-triviality
1. operations	$H^1(\Gamma, A) \cong A^{\beta_1}$, holonomy (Thms. 3, 4)	the operations are <i>known</i> ; $\beta_1 > 0$. If they are estimated, Theorem 7 returns and a fitted system supplies its own consistent field.
2. probability	$H^1(\mathcal{S}; \mathcal{F}_\alpha)$ one-dimensional, generated by entropy (Thm. 14)	the conditioning action is present; $I = \delta_t H$ measures its failure to be trivial.
3. outcomes	$\gamma \in \check{H}^1(\mathcal{U}, \mathcal{F}_{\check{U}})$, contextuality	the cover has <i>no</i> global context, and the support is sparse; any smoothing makes the model possibilistically extendable and γ vanishes (Prop. 10).

Remark 18 (The layers are a stratification, not a bridge). Each invariant is non-trivial in its own layer and zero in the others, and the reasons are the hypotheses themselves rather than any defect of construction. Layer 2’s site has a global joint—that is what makes the observable monoid have an absorbing element—and Theorem 27 then kills every trivially-acting coefficient system on it, which is where Layer 3’s coefficients would have to live. Layer 3’s site is built so that no global context exists, which is what permits $\gamma \neq 0$ and simultaneously denies Layer 2 the joint it requires. Layer 1 escapes both only by assuming its operations given rather than inferred.

So the pipeline runs forward and the invariants do not run back. Calling this a bridge would restore precisely the claim the computation refutes; the accurate statement is that the layers are separated by a theorem, and the separation is structural.

Remark 19 (Degree bookkeeping). γ is a class in $\check{H}^1$, being the connecting homomorphism of $0 \rightarrow \mathcal{F}_U \rightarrow \mathcal{F}_{\mathbb{Z}} \rightarrow \mathcal{F}|_U \rightarrow 0$ applied to a section of H^0 . Degree two enters this work only as the hypothetical target of a connecting map out of $H^1(\mathcal{F}_\alpha)$ —the object Theorem 27 shows to be zero. The two should not be conflated. Likewise the relation between Layers 2 and 3 is an incompatibility rather than a duality: a duality would supply a correspondence carrying one invariant to the other, and none exists.

Remark 20 (What the next step is). The stratification says where each invariant applies; it does not yet say what to compute when one actually works on a site and drives outcomes from it. That is the natural continuation: fix an information structure, carry probabilities on it, and ask which outcome-level quantities are determined and which are free—with Layer 1 available whenever the local operations are given, and Theorem 7 as the standing constraint that a single fitted model can never exhibit an obstruction to its own consistency.

7 Status of every claim

Proved here

- Theorem 3, holonomy obstruction, with the identification $H^1(\Gamma, A) \cong A^{\beta_1}$.

- Theorem 7, single-source global section.
- The three propositions of Section 2.

Computed, with controls

- Holonomy against brute force, $A = \mathbb{Z}/4096$, $\beta_1 = 2$: agreement on a consistent case and two inconsistent ones.
- The AMB obstruction on the 3×3 mod-3 parity family: 9/9 sections obstructed over $\mathbb{Z}$ and $\mathbb{Z}_3$, 0/9 over $\mathbb{Q}$ and $\mathbb{Z}_2$, with controls $(0, 0, 0)$ and $(1, 1, 1)$ at 0/9 and contextual fraction 0, 1, 0.
- Possibilistic fragility: γ from 9/9 to 0/27 at $\varepsilon = 10^{-9}$ while $\text{CF} = 1 - 3\varepsilon$ survives to $\varepsilon = 1/3$.
- Single-source vanishing on cyclic sliding-window covers of real corpora, with disjoint-cover and shuffled-data controls.
- $\omega \equiv 1$: the support-indicator bridge is constant on every family.

Open

Problem 21 (T1, site correspondence). Does an AMB measurement scenario determine an information structure in Vigneaux’s sense? His objects are observables with a conditioning action, not outcome sets. Unverified; if it fails, T2 is vacuous.

Problem 22 (T2, cover preservation). Is Φ a morphism of sites, sending $\mathcal{M}$ to a covering family? Depends on T1.

Problem 23 (T3, coefficient comparison — the blocked form). Exhibit a morphism of coefficient systems $\Psi : \mathcal{F}_{\mathbb{Z}} \rightarrow \mathcal{F}_{\alpha}$ intertwining the module structures. This is blocked, and the block is the one obstacle from which the others follow: $\mathcal{F}_{\alpha}$ ’s module structure *is* the conditioning action, $X \cdot f$ reweighting by $P(X=i)$, while $\mathcal{F}_{\mathbb{Z}}$ is the free abelian group on support sections and contains no probabilities. The required identity $\Psi(X \cdot v) = X \cdot \Psi(v)$ has an undefined left-hand side: not hard to satisfy, but not expressible. Four attempts to manufacture the missing side are recorded in Section 3, and all four failed.

Problem 24 (T3’, the extension formulation — the open form). Replace the search for a morphism by a search for an *extension*. A direct sum $\mathcal{F}_{\alpha} \oplus \mathcal{F}_{\mathbb{Z}}$, with the conditioning action on the first factor and trivially on the second, is a legitimate module, but cohomology commutes with direct sums,

$$H^k(\mathcal{F}_{\alpha} \oplus \mathcal{F}_{\mathbb{Z}}) = H^k(\mathcal{F}_{\alpha}) \oplus H^k(\mathcal{F}_{\mathbb{Z}}),$$

so the two classes sit side by side with no map between them. Carrying two objects in one bag does not relate them.

What relates them is a short exact sequence of coefficient systems

$$0 \longrightarrow \mathcal{F}_{\mathbb{Z}} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F}_{\alpha} \longrightarrow 0$$

that does *not* split. Its long exact sequence supplies a connecting homomorphism

$$\partial : H^k(\mathcal{F}_{\alpha}) \longrightarrow H^{k+1}(\mathcal{F}_{\mathbb{Z}}),$$

which is a map between the two cohomologies and requires no morphism $\mathcal{F}_{\mathbb{Z}} \rightarrow \mathcal{F}_{\alpha}$ — only an extension. The problem is therefore:

1. compute $\text{Ext}^1(\mathcal{F}_\alpha, \mathcal{F}_\mathbb{Z})$ over the information site;
2. exhibit a non-split extension, if one exists;
3. evaluate ∂ on the entropy class, the generator of $H^1(\mathcal{F}_\alpha)$, and determine whether its image in $H^2(\mathcal{F}_\mathbb{Z})$ is non-zero.

Three things make this the right target rather than a restatement of the blocked problem. It needs no morphism between the coefficient systems. The device is already in use on one side: the AMB obstruction γ is itself the connecting homomorphism of $0 \rightarrow \mathcal{F}_U \rightarrow \mathcal{F}_\mathbb{Z} \rightarrow \mathcal{F}|_U \rightarrow 0$. And there is a candidate for the extension data in the literature: Baudot and Bennequin’s identity

$$(\delta - \delta_t)F = ((S_0 - 1) \cdot F)$$

measures exactly the discrepancy between the acting and the non-acting module structures, which is what an extension between $\mathcal{F}_\alpha$ and a trivially-acting $\mathcal{F}_\mathbb{Z}$ would have to be built from.

Two constraints carry into the computation. First, the target group must be determined before ∂ can be evaluated on it, and Theorem 27 below shows what that determination gives. Second, the homological partner vanishes for a reason worth recording: $\mathcal{F}_\alpha$ is real-valued, hence flat over $\mathbb{Z}$, so $\text{Tor}_1(\mathcal{F}_\alpha, \mathcal{F}_\mathbb{Z}) = 0$. This is the same fact as the vanishing of γ over $\mathbb{Q}$ while it survives over $\mathbb{Z}$ and $\mathbb{Z}_3$: a mod- p contradiction is p -torsion, and $\mathbb{R}$ has none. Tor_1 becomes the interesting object only for a coefficient system deformed to carry torsion, where it would measure precisely the residue that γ detects.

Problem 25 (T6, parity criterion). A mod- p family on a cover is contradictory iff every variable lies in an *even* number of contexts and $\sum_C s_C \not\equiv 0 \pmod{p}$. Proof by XORing the constraints. Two proposed examples failed this test: a 3-cycle of 3-element contexts overlapping in single variables has 8 satisfying assignments, because three variables occur once and absorb the parity; and $(1, 1, 1)$ has 81, because $1+1+1 \equiv 0$. Ring dependence is a corollary: the class is seen exactly by rings that see the residue.

Problem 26 (Holonomy from estimated operations). If g is estimated from observations rather than known, the problem becomes an inverse one and Theorem 7 returns: a single fit supplies its own consistent field. Under what conditions is g identifiable, and when is a non-zero $[g]$ distinguishable from estimation error? This is the open question with the most empirical content.

Settled since: the target of ∂ vanishes

Theorem 27 (Vanishing on the trivially-acting complex). *Let $\mathcal{S}$ be an information structure whose observables form a monoid under joint with an absorbing element—the finest observable, the joint of everything—and let A be any abelian group with trivial action. Then $H^n(\mathcal{S}; A) = 0$ for all $n > 0$, and $H^0 = A$.*

Proof. With A discrete, a cochain measurable in P and local is constant in P , so $C^n = A^{|\mathcal{S}|^n}$ and the complex is the bar complex of the monoid with trivial coefficients, computing $H^*(B\mathcal{S}; A)$. A monoid with an absorbing element has contractible classifying space, so the cohomology vanishes in positive degrees for every coefficient group. $\square$

This was verified independently: for the join-semilattices of two and of three binary variables, the explicit bar complexes give $H^0 = \mathbb{Z}$ and $H^1 = H^2 = 0$ over $\mathbb{Q}$ and, by Smith normal form, over $\mathbb{Z}$ with no torsion. The control is H^0 , which Baudot–Bennequin derive independently (their eq. 28) and which the computation reproduces.

Corollary 28 (Problem 24 is settled negatively). *The connecting map ∂ of any extension of $\mathcal{F}_\alpha$ by a trivially-acting $\mathcal{F}_\mathbb{Z}$ lands in $H^{k+1}(\mathcal{F}_\mathbb{Z}) = 0$. No extension helps, split or not, and no choice of coefficients helps, since the vanishing holds for every abelian group and hence for any filtered or graded family of them.*

Remark 29 (The dichotomy, and what it says about the two theories). Either the coefficients act trivially, and Theorem 27 applies; or they do not, and the coefficient system is no longer $\mathcal{F}_\mathbb{Z}$, so the AMB class does not live in it. Either way there is no target.

The reason is structural and is sharper than the module argument of Section 5. An information structure presupposes that any set of variables has a joint—that is what produces the absorbing element. A measurement scenario presupposes that some sets do not: its contexts are maximal and there is no global context, which is exactly the condition that permits $\gamma \neq 0$. The two sites encode contradictory assumptions about joint measurability, so transporting either theory’s coefficients onto the other’s site destroys precisely the feature that made it non-trivial. This is not a failure to find a bridge; it is a demonstration that the two theories describe incompatible situations.

Closed by theorem, not to be reopened

- The p -adic bridge: $\mathbb{R}$ and $\mathbb{Q}_p$ are inequivalent completions (Ostrowski), and an order-preserving map constant on p -adic balls is constant.
- Branch cohomology with simplex coefficients: a rooted tree poset has an extremal object, so global sections is evaluation there, which is exact, so $H^k = 0$ for $k > 0$ with *every* coefficient system; truncation by mass does not help, since the root has mass 1 and survives every threshold. Separately, Δ^{V-1} is not a coefficient system.
- A map from β_1 of an incidence complex to γ : the former is constant across families, the latter is not.
- A non-trivial class for $I(X; Y \mid Z)$ obtained by restricting to smooth, boundary-regular or symmetric cochains: H and I share their $p \log p$ singularities, so no regularity condition separates them, and S_k invariance is vacuous in degree one.
- The identification of the boundary holonomy class with γ through a support-reduction map: that map is not additive.

References

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